### 1.1.1 What is Machine Learning?

Machine learning (ML) is the field of computer science/intelligent computing or branch of advance artificial intelligence (AI) that involved the development of self-learning algorithms to automatically extracts rules/knowledge from data/Big Data in order to make predictions/decisions. The data is basically historical records/past data points. Instead of requiring domain experts/humans to manually extract rules and build models from analysing Big Data, machine learning offers a more efficient alternative for gaining the knowledge in data to gradually improve the performance of predictive models and make data-driven decisions. It ables the machine/computer to learn from experience without being specifically programmed. ML build models based on historical data, known as *training data* for future prediction/decision making. ML is turning things (Big Data) into numbers and finding patterns in those numbers. ML is used in many real-life applications, e.g. internet search engines, email filters to sort out spam, detecting intrusions/unusual transactions, image classification etc.

**Arthur Samuel (1959)** Machine Learning: Field of study that gives computers the ability to learn without being explicitly programmed.

**Tom Mitchell (1998)** Well-posed Learning Problem: A computer program is said to learn from experience **E** with respect to some task **T** and some performance measure **P**, if its performance on **T**, as measured by **P**, improves with experience **E**.

**Dewan Farid (2021)** Machine Learning is the process of extracting rules from Big Data for knowledge mining.

### 1.1.2 What is Data Mining?

Data mining is also known as *Knowledge Discovery from Data*, or *KDD* for short, which turns big data into knowledge. It's the process of extracting knowledge and uncovering hidden patterns from Big Data to understand the current and future trends. The main object of data mining is to extract patterns (known and unknown) from Big Data. The following definitions are found from different literatures.

1. It's the process of analysing data from different perspectives and summarising it into useful information.

2. It's the process of finding hidden information and patterns in a huge database.

3. It's the extraction of implicit, previously unknown, and potentially useful information from data.

### 1.1.3 Why Machine Learning and Data Mining?

Can we think of all the rules in complex real-life Big Data problems with long lists of rules? Also, environments are continually changing in real-life situations that's why we need machine learning and data mining. ML can adapt ("learn") to new scenarios when the concepts are changing over the time. The terms machine learning and data mining are commonly confused, as they often employ the same methods and overlap significantly. They can be roughly defined as follows:

- Machine learning focuses on prediction, based on known properties learned from the training data.
- Data mining focuses on the discovery of unknown properties in the data.

**Artificial Intelligence (AI)** The effort to automate intellectual tasks that normally performed by humans. It enables computers to mimic human behaviour. The basic idea is to create intelligent machines that work and react like a human brain.

**Deep Learning** Subset of machine learning in which multilayered artificial neural networks (ANN) learn from Big Data.

**Big Data** Large volume of data with variety and velocity.

**Data Science** Data Science is the field of study that combines knowledge of artificial intelligence (AI), machine learning (ML), data mining (DM) to extract knowledge and insights from structured and unstructured data.

**Data Warehouse** A Data Warehousing (DW) is the process of compiling, managing and organising data from several sources into one common database, whereas data mining refers to the process of extracting useful data from the databases.

### 1.1.4 Data, Information, Knowledge

**Data** It's a collection of text, numbers, symbols, sound, picture or any recorded facts in raw or unorganised form that can be processed by a computer. Data can be collected from different sources, e.g. scientific data, medical data, demographic data, financial data, and marking data. It's really important to have good quality data for Machine Learning research and modelling. Engendering data is a challenging and costly process, which is the most important part of all Data Analytics, Data Mining, and Big Data research.

**Information** Processed data is call information. Data that has been processed, e.g. grouped, normally by a computer, to give it meaning and make it interpretable. The patterns, associations or relationships in data can provide information.

**Knowledge** Processed information is call knowledge that is the understanding of information such as how to solve problems. Information can be converted into knowledge about historical patterns and future trends.

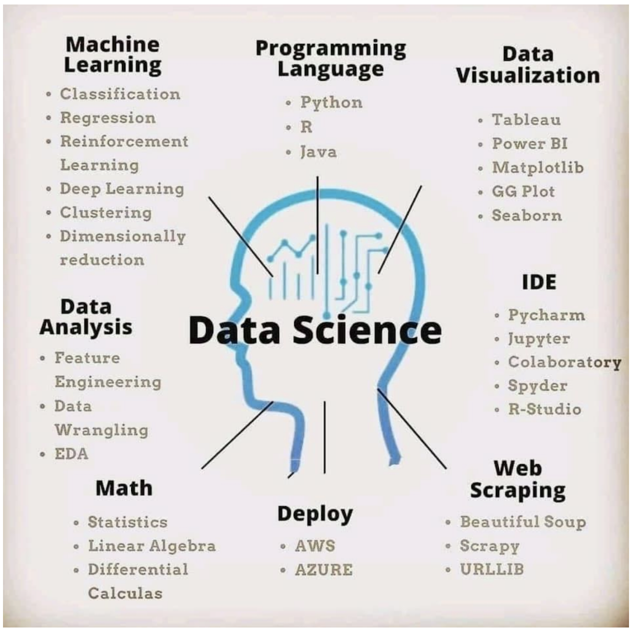


Figure 1.1: Data Science.

### 1.1.5 Types of Data for Mining

Data can be classified into two categories: **Unstructured Data** and **Structured Data**. Unstructured data have no rigid structure, e.g. images, video, natural language text, and speech etc. Structured data is like a table, which can be classified as follow:

1. Nominal/Categorical (e.g. marital status, eye colour, political party etc)

2. Numerical (discrete: number of children, defects per hour; and continuous: weight, sales, sensor stream data)

3. Ordinal data, which has order but the distance between values is unknown. E.g. how would you rate your health from 1-5? where, 1 being poor, 5 being healthy.

4. Time-series Data (Data across time), e.g. the historical scale values of gold from 2012-2018.

Real world data are generally incomplete, noisy, and inconsistent. ***Low quality data will lead to low quality mining results***. Generally, big data contains errors, missing attribute values, and lacking certain attributes of interest. Measure for data quality is a multidimensional view, which follows:

1. Accuracy: correct or wrong, accurate or not.

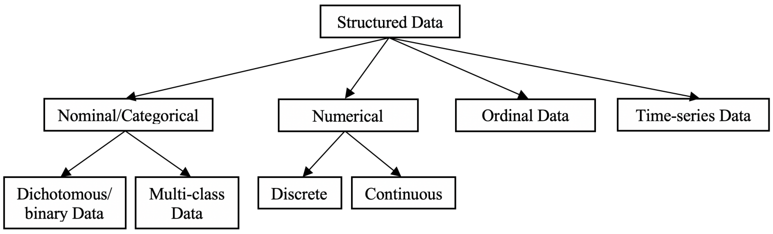


Figure 1.2: Types of structured data.

2. Completeness: not recorded, unavailable.
3. Consistency: some modified but some not.
4. Timeliness: timely update?
5. Believability: how trustable the data are correct?
6. interpretability: how easily the data can be understood?

## 1.2 Learning Types in Machine Learning

The general idea of machine learning is to develop concepts from historical data for acquiring knowledge through experience. There are different types/problems of learning in the field of machine learning that are listed below.

### 1.2.1 Supervised Learning

Supervised learning is the process of building models when input variables, $X$, called features and output variable, $y$ called label or a class employ algorithms to learn mapping function $y = f(X)$ from input/data to output/class. The goal is to map the function so well that the new input data/instance $x'$ can be classified correctly. In supervised learning, we have data and labels. The ML model tries to learn the relationship between data and labels. Supervised learning can be bifurcated into the following tasks:

***1.2.1.1 Regression*** It consists of mathematical methods that allow us to predict the numerical/real/continuous class-value/label of output variable $y$ based on the value of one or more predictor/input variables $X_i$. Linear regression is one of the most common form of regression analysis that used in prediction and forecasting.

***1.2.1.2 Classification*** It refers to the predictive modelling problems where the class-values/labels of output variable are categorical/nominal/discrete, i.g. classify if an email is spam or not. There are three types of classification problems:

1. Binary-class classification.

2. Multi-class classification.

3. Multi-label classification (e.g. what items does this photo contain? What topics is this YouTube video about?).

### 1.2.2 Unsupervised Learning

Unsupervised learning also known as clustering, is used to grouping unlabelled data (when only have input data, $\mathbb{X}$ and no corresponding class-values) for finding patterns/similarities and interesting structure in the data. The goal for unsupervised learning is to model the underlying structure or distribution in the data in order to learn more about the data. Therefore, in unsupervised learning we have only data but no labels. The ML model tries to find the pattern in data without something to reference on. Clustering is commonly used for dimensionality reduction, e.g. PCA (Principal Component Analysis).

### 1.2.3 Semi-supervised learning

Semi-supervised learning is the process of learning models from semi-supervised data where we have both labeled and unlabelled instances.

### 1.2.4 Reinforcement Learning

Reinforcement learning is an agent (algorithm) performs actions in an environment and is rewarded or penalised based on the whether the actions were favourable or not, Fig 1.3. In reinforcement learning, there's no training data and the algorithm works on a rewards-based system. The agent selects an action in an environment that will lead to rewards, e.g. creating a game.

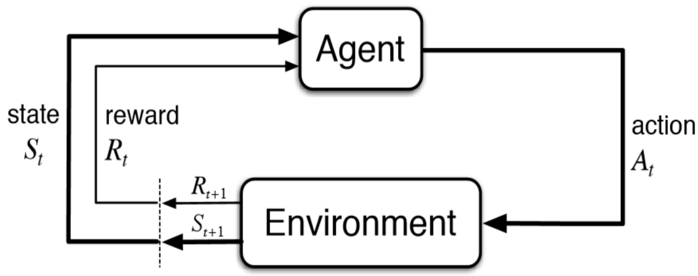


Figure 1.3: Reinforcement Learning.

### 1.2.5 Transfer Learning

Transfer learning takes knowledge from one/existing model and use it in your own/other model.

### 1.2.6 Active Learning

Active learning is a machine learning approach that lets users play an active role in the learning process. An active learning approach can ask a user (e.g., a domain expert) to label an instance, which may be from a set of unlabelled instances.

### 1.2.7 Sequence to Sequence

Sequence to sequence (Seq2seq) is one of the machine learning techniques used for language processing. Applications include language translation, image captioning, conversational models and text summarisation, e.g. given a sequence of English text, translate it into France.

## 1.3 The Weather Problem

To illustrate an example, we consider a small dataset in Table 1.1 described by four attributes namely Outlook, Temperature, Humidity, and Wind, which represent the weather condition of a particular day. Each attribute has several unique attribute values. The Play column in Table 1.1 represents the class category of each instance. It indicates whether a particular weather condition is suitable or not for playing tennis.

Table 1.1: Weather Data

| **Outlook** | **Temperature** | **Humidity** | **Wind** | **Play** |
|---|---|---|---|---|
| Sunny | Hot | High | Weak | No |
| Sunny | Hot | High | Strong | No |
| Overcast | Hot | High | Weak | Yes |
| Rain | Mild | High | Weak | Yes |
| Rain | Cool | Normal | Weak | Yes |
| Rain | Cool | Normal | Strong | No |
| Overcast | Cool | Normal | Strong | Yes |
| Sunny | Mild | High | Weak | No |
| Sunny | Cool | Normal | Weak | Yes |
| Rain | Mild | Normal | Weak | Yes |
| Sunny | Mild | Normal | Strong | Yes |
| Overcast | Mild | High | Strong | Yes |
| Overcast | Hot | Normal | Weak | Yes |
| Rain | Mild | High | Strong | No |

A set of rules learned from the Table 1.1.

1. If Outlook = Sunny and Humidity = High then Play = No
2. If Outlook = Sunny and Humidity = Normal then Play = Yes
3. If Outlook = Overcast then Play = Yes
4. If Outlook = Rain and Wind = Strong then Play = No
5. If Outlook = Rain and Wind = Weak then Play = Yes

In the slightly more complex form shown in Table 1.2, two of the attributes - temperature and humidity have numeric values.

Table 1.2: Weather Data with Some Numeric Attributes

| Outlook | Temperature | Humidity | Wind | Play |
|---|---|---|---|---|
| Sunny | 85 | 85 | Weak | No |
| Sunny | 80 | 90 | Strong | No |
| Overcast | 83 | 86 | Weak | Yes |
| Rain | 70 | 96 | Weak | Yes |
| Rain | 68 | 80 | Weak | Yes |
| Rain | 65 | 70 | Strong | No |
| Overcast | 64 | 65 | Strong | Yes |
| Sunny | 72 | 95 | Weak | No |
| Sunny | 69 | 70 | Weak | Yes |
| Rain | 75 | 80 | Weak | Yes |
| Sunny | 75 | 70 | Strong | Yes |
| Overcast | 72 | 90 | Strong | Yes |
| Overcast | 81 | 75 | Weak | Yes |
| Rain | 71 | 91 | Strong | No |

A set of rules learned from the Table 1.2.

1. If Outlook = Sunny and Humidity $> 75$ then Play = No
2. If Outlook = Sunny and Humidity $\leq 75$ then Play = Yes
3. If Outlook = Overcast then Play = Yes
4. If Outlook = Rain and Wind = Strong then Play = No
5. If Outlook = Rain and Wind = Weak then Play = Yes

### 1.3.1 Input: Concepts, Instances, and Attribute

The input takes the form of *concepts*, *instances*, and *attributes*. We call the thing that is to be learned a *concept description*. Each instance is characterised by the values of attributes that measure different aspects of the instance. There are many different types of attributes, although typical data mining schemes deal only with numeric and nominal, or categorical attributes.

**Concept** is the thing to be learned.

**Concept description** is the output produced by a learning scheme or classifier.

**Instances** are the things that are to be classified or associated or clustered. Each dataset is represented as a matrix of instances versus attributes, which in database terms is a single relation, or a *flat file*.

**Attribute** is a data field, representing a characteristic or feature of a data object. The nouns *attribute, dimension, feature*, and *variable* are often us interchangeably in the literature. The value of an attribute for a particular instance is a measurement of the quantity to which the attribute refers.

## 1.4 Machine Learning Process

Learning is the concept of estimating the model parameters so the predictions are consistent with true labels. There are six steps in the machine learning process: (1) data collection, (2) data preparation, (3) training model, (4) testing model, (5) deploying model, and (6) retrain model that's shown in Fig. 1.4.

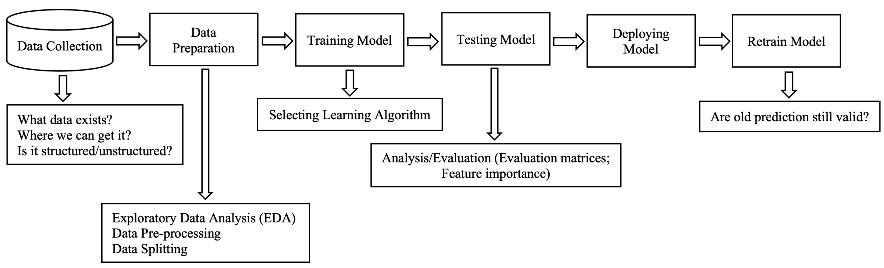


Figure 1.4: Machine Learning Process.

## 1.5 Data Collection and Preparation

To collect the data we need to ask the following questions:

- What kind of problems are we trying to solve?
- What data sources already exist?
- What privacy concerns are there?

- Is the data public?
- Where should we store the data?

Data Preparation performs the following tasks:

- Exploratory Data Analysis (EDA), i.g. learning about the data, understand and know the data;
- Data Preprocessing (preparing data to be modelled)
- Data Splitting

### 1.5.1 Exploratory Data Analysis (EDA)

Learning about the data/understand and know the data:

- What are the feature variables (input) and the target variable (output)?
- What's the kind of data? e.g. structured/unstructured
- Are there missing values?
- Where are the outliers? how many of them are there? why are they there?

### 1.5.2 Data Preparation

- Feature Imputation (filling missing values)
- Feature Encoding (turning values into numbers)
- Feature Normalisation (scaling) or Standardisation
- Feature Engineering: transform data into (potentially) more meaningful representation by adding in domain knowledge.
- Feature Selection (selecting most valuable features)
- Dealing with Imbalances

### 1.5.3 Data Splitting

- Training Set (usually 70-80%)
- Validation Set (typically 10-15%): Model hyper-parameters are turned on this;
- Testing Set (10-15%)